CHARCOAT

CharCoat FM
(4 Hour Fire Rated Mortar)

TDS

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PURPOSE/CHANGES	DESIGN
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PRODUCT DESCRIPTION

CharCoat FM Mortar is a dry white powder consisting of gypsum and perlite.

When mixed with water, the compounds form a highly thermally insulating 4 hour fire sealing compound to prevent the spread of fire and smoke through openings in fire rated walls and floors, including openings formed around building service penetrations.

CharCoat FM will also maintain the acoustic design performance in walls and floors.

CharCoat FM expands approx. 1% by hydraulic action during curing ensuring a very tight seal around the service penetrations and the surrounding opening apertures.

CharCoat FM is easy to sand or drill. The compound dries to an off-white colour which may be painted.

PROPERTIES

- Up to 4 hours Fire Rating
- Classified in walls and floors of concrete, brick, gypsum etc.
- Suitable for cables, bundled cables, cable racks, cable trays, steel, copper, alupex, plastic pipes and air ventilation ducts
- Simple to apply leaving a very smooth finish
- High degree of mechanical resistance; the seal is load bearing without reinforcement
- No priming necessary prior to application in most building material substrates however metal services in contact with the seal must be corrosion protected
- Suitable for most surfaces, included concrete, bricks, Leca, steel, plastic etc, but not suitable to fittng of doors or in service openings that involve movement
- The product is cerfied for use in walls and floors
- Fully set within 1 hour
- The fire performance specificaon of the compound has been derived when the seal has been left to cure for 1 month
- Nearly unlimited storage time

SOUND INSULATION

Description	Sound Reduction
Single sided Cast ≥ 50mm on stone wool board	64 dB
Single sided Cast ≥ 100mm without board	64 dB
Double sided Cast ≥ 25mm on stone wool board	64 dB
Double sided Cast ≥ 50mm without board	64 dB

CharCoat FM has been tested at BM Trada (UKAS accredited); according to EN ISO 10140-2:2010.

RESISTANCE TO FIRE

Construction	Description	Classification
Flexible walls comprise gypsum, masonry, aerated concrete or concrete	Up to 2400mm wide x 1200mm high blank seals with double sided 25mm CharCoat FS Mortar on 25mm cast board	EI 120 (E 120)
Rigid walls comprise masonry, aerated concrete or concrete, within walls or between the head of walls and the soffit of floor slabs	Up to 2400mm wide by 1200mm high blank seal with single sided 50mm CHarCoat FS Mortar on 50mm cast board	EI 120 (E 180)
	Up to 2400mm wide by 1200mm high blank seal with single sided 100mm CharCoat FS Mortar	EI 240 (E 240)
Rigid floors comprise aerated concrete or concrete within floors or between floors and walls	Up to 2400 mm by 1200 mm blank seal with 50mm CharCoat FS Mortar on top of 50mm cast board	EI 180 (E 180)
	Up to 2400 mm by 1200 mm blank seal casted with 100mm CharCoat FS Mortar	EI 240 (E 240)

The cast board comprise stone wool with density ≥ 150kg/m³. Please read the Installation Instructions before usage.

LOAD BEARING PROPERTIES (FLOORS)

CharCoat FM has been subject to concentrated load and impact-tests in floors. The tests were conducted on the minimum allowed cast depth of 100mm.

According to the loading limits in the table below, reinforcement is not necessary, however it is highly recommended that the edges of the aperture are brushed free of any dust or loose particles and that any contamination is washed away using clean water. Moistening the edges well before casting will improve adhesion.

CharCoat FM should not be cast in surface treated concrete. The mortar must be mixed to a thick but fluid mass at a rate of approx. 2 parts of powder to 1 part water. Maximum load bearing performance will be achieved 28 days after casting.

TEST RESULTS:

Test in 1500x1000mm frame	Results
Soft body impact, serviceability	500Nm
Soft body impact, safety in use	700Nm
Hard body impact, serviceability	6Nm
Hard body impact, safety in use	10Nm
Concentrated load to ETAG 26-2	15kN

EMISSION DATA (INDOOR AIR QUALITY)

Compound	Emmisions rate after 3 days	Emmisions rate after 4 days
TVOC	12 µg/m³	< 5 µg/m³
TSVOC	n.d.	< 5 µg/m³
VOC w/o NIK	n.d.	< 5 µg/m³
R Value	n.d.	< 1
Formaldehyde	7.1 µg/m³	n.d.
Acetaldehyde	< 3 µg/m³	n.d.
Sum for+ace	< 0.006 ppm	n.d.
Carcinogenic	< 1 µg/m³	< 1 µg/m³
n.d. or < means not detected		

CharCoat FM complies with the requirements of GEV and the results correspond to the EMICODE emission class EC 1plus which is the best possible environmental and indoor hygiene health protection mark.

CURING TIMES

Application	Temperature	Cure Time
For filler 3.5 to 1 mix	0 °C	19 minutes
	10 °C	18 minutes
	20 °C	17 minutes
	30 °C	16 minutes
	40 °C	15 minutes
For casting 2 to 1 mix	0 °C	40 minutes
	10 °C	35 minutes
	20 °C	30 minutes
	30 °C	25 minutes
	40 °C	20 minutes

CharCoat FM can be mixed with an electric mixer for 90 seconds at 750 rpm with a 100mm diameter paddle. Note the greater the sheer/agitation generated in the mixing process the quicker the mortar will set.

CharCoat FM is designed to be a quick curing system for professional installers where fast application times is of the highest importance. For slower cure, a retardant can be added to the dry mortar powder (sold separately).

TECHNICAL DATA

Condition	Powder ready for mixing with water
Product consumption at 2:1 mix	Approx. 4 bags per m2 @ 100mm depth
Dry density	About 900 kg/m3 after full cure
Flash point	None. Class A1 according to EN 13501-1
Hardened	Less than 1 hour depending on the local climate
Totally hardened	Up to 30 days depending on thickness and temp.
Flexibility	None
Thermal conduct	0.051 W/mK
Storage	3 years for unopened bags in dry places with storage temperatures between 5°C and 30°C
Compatibility Limitations	Suitable for use with most materials, but should not be used in direct contact with metals that may corrode
Colour	Off white
Packaging	20kg Bags Bags: 50 on the pallet, equals approx. 1000 kg
Warranty	25 Years / Product only. Please refer to our full Terms and Conditions of sale.

No liability can be accepted for the information provided in this document although it is published in good faith and believed to be correct. CharCoat Passive Fire Protection reserves the right to alter product specifications without prior notice, in line with Company policy of continuous development and improvement.

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